

APOORVA DEEP SINGH

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EDUCATION

Uttaranchal University, Dehradun – B.Tech Computer Science & Engineering

2023 – 2027

CGPA: 8.0/10 (Till 6th Semester)

DAV Public School, Patna – Senior Secondary (XII: 65% | X: 73%)

2021 – 2023

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, HTML, CSS, SQL, Java
Frameworks: Node.js, Express.js, React, FastAPI, MongoDB, PostgreSQL, SQLite, REST APIs
AI/ML & Systems: RAG Pipelines, Vector Databases, sentence-transformers, LLM Integration, System Design, Microservices
Tools & Concepts: Git, GitHub, Docker, VS Code, Postman, DSA, OOP, Agile; **131+ LeetCode problems solved**

PROJECTS

AI Career Engine | Python, FastAPI, RAG, Vector DB, sentence-transformers, OpenAI API

- Engineered production-grade career intelligence platform using custom RAG pipeline; delivers personalized learning roadmaps, ranked skills, job roles, and salary insights from semantic search across 5 vector collections.
- Implemented 384-dimensional embeddings (MiniLM) enabling meaning-aware retrieval; designed async FastAPI backend with parallel multi-collection retrieval achieving **sub-100ms query latency**.
- Integrated OpenAI/Groq LLMs for structured career report generation; built as part of AI Engineering Internship Evaluation (Endee.io) demonstrating end-to-end system design.

Mess-Wise — Smart Mess Management System | TypeScript, Node.js, React, SQLite, RBAC

- Built full-stack SPA automating hostel cafeteria operations; reduced manual meal tracking overhead by tracking attendance, rewards, and penalties across **100+ daily users**.
- Implemented role-based access control (Admin/User), RESTful APIs, and SQLite persistence for meal consumption logging; designed responsive dashboard with real-time analytics and food wastage metrics.
- Architected optional IoT integration layer (Raspberry Pi simulation) for hardware-driven meal scan automation, enabling future scalability to hardware ecosystem.

AirSync — Cross-Platform File Transfer System | Python, Networking, Multi-threading

- Developed peer-to-peer file transfer utility supporting Windows, Linux, macOS; eliminated third-party cloud dependency for local network file sharing with **secure data transfer protocols**.
- Optimized multi-threaded architecture achieving OS-agnostic compatibility without external dependencies; reduced transfer latency through parallel chunk processing.

Hope — Mini Compiler | C++, Compiler Design, Tokenization

- Implemented foundational compiler phases: lexical analysis, syntax parsing, and intermediate code generation; validated core compiler design concepts including tokenization and AST construction.

JobAce — Resume Analyzer | Python, NLP, Data Analysis, REST APIs

- Created resume-job description matching engine using skill extraction and keyword density analysis; provides skill gap analysis and optimization recommendations via REST API.

ACHIEVEMENTS & RECOGNITION

LeetCode Performance – Solved 131+ DSA problems; ranked in top 15% for consistency

GitHub Contributions – 5 full-stack projects with 100+ commits; active open-source contributor

AI Engineering Evaluation – Recognized at Endee.io for production-grade system design in AI Career Engine project

CERTIFICATIONS

Full Stack Developer Bootcamp (GeeksforGeeks) – Node.js, React, Database Design | Agile Methodology (HP) – Scrum, Sprint Planning
Backend Development Certification (PW Skills) – REST APIs, Database Optimization | ChatGPT & AI Integration – LLM Fundamentals, Prompt Engineering

CORE COMPETENCIES

Backend Development: RESTful APIs, Async Programming, Database Design, Microservices Architecture, Real-time Systems

AI/ML Systems: RAG Implementation, Vector Embeddings, LLM Integration, Semantic Search, System Optimization

Full-Stack Development: Single Page Applications, Responsive UI/UX, State Management, Real-time Communication (WebSockets)

System Design: Scalability, Performance Optimization, Multi-threading, Concurrency, Cloud-Ready Architecture

PROFESSIONAL INTERESTS & FOCUS

Specializations: AI-driven Backend Systems • RAG Pipelines & Vector Databases • Microservices Architecture • System Design at Scale | Current Learning: Advanced Distributed Systems • AI Agent Development • Real-time Data Processing • LLM Fine-tuning
Community: Actively contribute to GitHub • Mentor junior developers in DSA • Regular competitive programming practice • Exploring open-source opportunities in AI/ML space

EXPERTISE HIGHLIGHTS

API Development: Designed and deployed 15+ RESTful APIs handling 1000+ concurrent requests; optimized response times using caching and database indexing strategies

Database Optimization: Normalized schema designs; implemented query optimization reducing latency by 40%; worked with MongoDB aggregation pipelines and PostgreSQL indexing

Frontend-Backend Integration: Built seamless React-Node.js communication; implemented WebSocket for real-time updates; state management with Redux and Context API

Deployment & DevOps: Docker containerization for microservices; GitHub Actions for CI/CD pipelines; environment configuration management across dev/staging/prod

Performance & Scalability: Async/await optimization; multi-threading implementation; load testing and horizontal scaling strategies; caching layers (Redis simulation)

TECHNICAL APPROACH & METHODOLOGY

Development Process: Agile sprint cycles • Code review best practices • Test-driven development (TDD) • Documentation standards • Version control (Git workflows)
Problem Solving: Object-oriented design patterns • SOLID principles • Algorithm optimization (Big O analysis) • System design interviews preparation • Architectural decision trade-offs
Code Quality: Writing clean, maintainable code • Reducing technical debt • Code refactoring • Performance profiling • Security best practices (data validation, encryption, authentication)